A Computable CG-Corrected Residual Certificate for Learned Elliptic Solvers

Error-Certified Neural Operators Project

***Abstract*—We present a computable a posteriori certificate for the *discrete* error of a learned surrogate of the elliptic system $A_h u_h = f$, obtained from $-\nabla \cdot (a\nabla u) = f$. Given a candidate $\hat{u}$, the exact identity $e = u_h - \hat{u} = A_h^{-1} r$, $r = f - A_h\hat{u}$, yields the classical bound $\|e\|_2 \le \|r\|_2/\lambda_{\min}(A_h)$, which is safe but ignores the structure of r and becomes severely conservative for heterogeneous coefficients. We instead compute z by applying conjugate gradients to $A_h z = r$ and form $B_{\mathrm{CG}} = \|z\|_2 + \|r - A_h z\|_2/\alpha$ with a coercivity-based $\alpha = a_{\min}\lambda_{\min}(-\Delta_h) \le \lambda_{\min}(A_h)$. We prove $B_{\mathrm{CG}} \ge \|e\|_2$ for arbitrary z, and we state explicitly the scope of that guarantee: it bounds the algebraic error with respect to u_h and is silent about discretization error, which no residual-based estimator can detect. In experiments computed from scratch for this manuscript (200 coefficient fields per contrast, eight contrasts, and 80 real trained-FNO predictions per test split, including two out-of-distribution splits) we measure median effectivity 1.000000 for B_{CG} against 316–1429 for the global bound. Because the certified quantity is itself an algebraic residual, near-unit effectivity reflects optimality on the declared target rather than a discovered property; the practically meaningful consequences are a measured acceptance policy—at tolerance $\tau = 5 \times 10^{-2}$ the certificate accepts 100% of instances where the global bound accepts 0%, both with zero false negatives—and a cost of 3.7–33 ms per instance for $15 \le n \le 127$. All code, raw JSON outputs, and the frozen configuration are public.**

***Index Terms*—error certification, neural operators, a posteriori error estimation, conjugate gradients, selective verification, elliptic PDEs**

I. Introduction

Learned surrogates such as the Fourier Neural Operator (FNO) [8], [7] and DeepONet [9] promise order-of-magnitude speedups on parametric PDEs, and are now reviewed as a computational tool in their own right [2]. For deployment in scientific workflows, however, a mean accuracy figure is the wrong currency: a user needs to know, for *this* instance, whether the prediction is accurate enough to act on. Statistical uncertainty quantification—ensembles, Bayesian heads, conformal prediction—answers a distributional question and does not yield a per-instance deterministic bound [15], [13].

A parallel line of work pursues certificates rather than distributions: rigorous a posteriori error control for PDE approximations [5], [12], and formal verification of neural surrogates by compiling them into SMT solvers [3]. Our contribution sits in the classical a posteriori tradition, specialized to the discrete error of a learned solver and paired with a fallback policy. We also pursue a complementary activity to certification: residual-based error *correctors* for neural operators improve accuracy but provide no guarantee [4].

For the discrete elliptic system the residual identity is exact and the classical bound is immediate. The question is whether it can be made sharp without losing rigor. We answer yes, and we are careful about what that means.

a) Contributions.: (i) A computable certificate $B_{\mathrm{CG}} = \|z\|_2 + \|r - A_h z\|_2/\alpha$ for $\|u_h - \hat{u}\|_2$, with a proof (Thm. 1) that requires only $A_h \succ 0$ and $\alpha \le \lambda_{\min}(A_h)$. (ii) A rigorous construction of α from problem data via coercivity, with no spectral estimation and hence no assumption that a numerical eigensolver is conservative. (iii) An explicit statement of the certificate's *scope*, including the fact that near-unit effectivity is a consequence of the target being an algebraic residual rather than an empirical discovery. (iv) Experiments regenerated from scratch for this manuscript, covering eight coefficient contrasts, three test splits with a trained FNO including two OOD splits, a tolerance sweep, and a cost-scaling study; all numbers are computed from committed JSON. (v) A measured selective-verification policy with an acceptance/fallback trade-off, and an explicit account of what this construction cannot certify.

II. Setting and Notation

Let $\Omega = (0,1)^d$, and consider

$$-\nabla \cdot (a(x)\nabla u) = f \quad \text{in } \Omega, \qquad u = 0 \quad \text{on } \partial\Omega, \tag{1}$$

with $0 < a_{\min} \le a(x) \le a_{\max} < \infty$. A conforming discretization with homogeneous Dirichlet boundary conditions yields the symmetric positive definite (SPD) system

$$A_h u_h = f, \qquad A_h \in \mathbb{R}^{n\times n}, \quad A_h \succ 0, \tag{2}$$

where n is the number of interior unknowns. Given any candidate $\hat{u} \in \mathbb{R}^n$, we define the residual $r = f - A_h\hat{u}$ and the error $e = u_h - \hat{u}$. Because A_h is invertible, the following identity is exact rather than an estimate:

$$e = A_h^{-1} r. \tag{3}$$

Throughout, $\|\cdot\|$ is the Euclidean norm. We certify $\|e\|$; the relationship to the error in the exact solution u is discussed in Sec. V.

III. The Classical Bound and Why It Is Loose

Lemma 1 (Global residual bound)**.** *If $\alpha > 0$ with $\alpha \le \lambda_{\min}(A_h)$, then $\|e\| \le \|r\|/\alpha$.*

Proof. By (3) and $\|A_h^{-1}\| = \lambda_{\min}(A_h)^{-1}$, $\|e\| \le \lambda_{\min}(A_h)^{-1}\|r\| \le \alpha^{-1}\|r\|$. □

Lemma 1 is attained when r lies entirely in the eigenspace associated with $\lambda_{\min}$, so no improvement is possible while the only information used is the pair $(\|r\|, \alpha)$. The looseness is

not a defect of the proof; it reflects that the bound charges every residual component at the worst spectral rate. For the heterogeneous coefficients typical of subsurface or composite-media models, the ratio $a_{\max}/a_{\min}$ enters $\lambda_{\min}(A_h)$ and the resulting factor is large.

Remark 1 (Measured looseness)**.** In our experiments (Sec. VII) the median of $\|r\|/\|e\|$ ranges from 316 at contrast 1.5 to 1429 at contrast 10. The bound is safe and, at these ratios, not useful for deciding whether an individual prediction is accurate.

IV. The CG-Corrected Certificate

Theorem 1 (CG-corrected certificate)**.** *Let $A_h \succ 0$, let $r = f - A_h\hat{u}$, and let $\alpha > 0$ satisfy $\alpha \le \lambda_{\min}(A_h)$. Then for every $z \in \mathbb{R}^n$,*

$$\|u_h - \hat{u}\| \;\le\; B_{\mathrm{CG}}(z) := \|z\| + \frac{\|r - A_h z\|}{\alpha}. \tag{4}$$

In particular, (4) *holds with z any iterate of conjugate gradients applied to $A_h z = r$, at any stopping point.*

Proof. Since $A_h z = A_h(e - z)$ by (3), the error splits as $e = z + A_h^{-1}(r - A_h z)$. Applying the triangle inequality and then Lemma 1 to the residual $r - A_h z$ gives $\|e\| \le \|z\| + \|r - A_h z\|/\alpha$. □

Two consequences matter. First, the bound is valid for *every* z, so a solver may stop early—after one iteration or a thousand—and the result remains a certificate; tightness and cost are traded against one another without any risk of losing validity. Second, the statement contains no assertion about the quality of z, only about its residual, which is directly computable.

Proposition 1 (Acceptance soundness)**.** *Under Assumption 1 and Theorem 1, a policy that accepts $\hat{u}$ whenever $B_{\mathrm{CG}} \le \tau$ satisfies:* $\mathrm{accept} \Rightarrow \|u_h - \hat{u}\| \le \tau$. *Hence the policy has zero false negatives.*

A. Constructing α without spectral estimation

A certificate is only as trustworthy as α. We do not estimate eigenvalues. For the standard finite-difference discretization of (1) with $a \ge a_{\min}$ and homogeneous Dirichlet data, the discrete energy satisfies, for all v,

$$v^{\mathsf{T}} A_h v \;\ge\; a_{\min}\, v^{\mathsf{T}}(-\Delta_h)v, \tag{5}$$

so by the min–max principle

$$\lambda_{\min}(A_h) \;\ge\; a_{\min}\,\lambda_{\min}(-\Delta_h), \qquad \alpha := a_{\min}\,\lambda_{\min}(-\Delta_h). \tag{6}$$

For uniform grids with Dirichlet boundary conditions $\lambda_{\min}(-\Delta_h)$ is known in closed form, equal to $4h^{-2}\sin^2(\pi h/2)$ in one dimension and $8h^{-2}\sin^2(\pi h/2)$ in two. The bound (6) is thus a deterministic function of the problem data and the mesh. We stress that a numerically computed Lanczos or Gershgorin estimate of $\lambda_{\min}$ would not be admissible without a guarantee that the estimate lies below the true value; using a coercivity bound avoids that question entirely, at the price of conservativeness that Thm. 1 then recovers.

Assumption 1 (Validity conditions)**.** (i) $a(x) \ge a_{\min} > 0$ on Ω; (ii) $A_h \succ 0$ arises from the conforming discretization of (1) with homogeneous Dirichlet conditions, so that (6) holds; (iii) the coefficient lower bound $a_{\min}$ used in (6) is a true lower bound on a; (iv) all linear algebra is carried out in IEEE double precision and $r - A_h z$ is evaluated explicitly rather than through the recurrence; (v) the guarantee is with respect to u_h.

Remark 2 (Floating point)**.** We do not apply outward rounding. As in classical computable CG bounds [10], [11], [6], the returned value is the bound evaluated on the computed z and on the explicitly recomputed residual $r - A_h z$. Since that residual is recomputed from z rather than accumulated through the iteration, the quantity entering (4) is the true residual of the returned vector up to rounding of the matrix-vector product. Adding a validated guard band would convert this into a machine-checkable certificate; we regard it as future work.

V. Scope of the Guarantee

We state the limits of (4) as precisely as we state its validity, because the central numerical result of this paper is easily misread.

a) The certificate is optimal on its target.: The quantity bounded is $\|u_h - \hat{u}\| = \|A_h^{-1} r\|$. As CG converges, $z \to A_h^{-1} r$ and $r - A_h z \to 0$, so $B_{\mathrm{CG}}(z) \to \|A_h^{-1} r\| = \|e\|$. The near-unit effectivity we measure is therefore a property of the target, not a discovery about neural operators. The same observation explains why no estimator of this form can be improved further on this quantity.

b) Discretization error is not included.: The exact solution satisfies

$$\|u - \hat{u}\| \;\le\; \underbrace{\|u - u_h\|}_{\text{discretization, not certified}} \;+\; \underbrace{\|u_h - \hat{u}\|}_{\text{certified by Thm. 1}}, \tag{7}$$

and only the second term is bounded. Estimating the first is a separate, well-understood a posteriori finite-element problem [1], [14].

c) Mis-specification is invisible.: If a surrogate is wrong because it was trained against, or evaluated with, a different operator or boundary data than the one used to form A_h, then A_h is not the operator whose error the user cares about, and B_{CG} remains small while the prediction is wrong. This failure mode is a property of any algebraic residual estimator and is noted explicitly in the recent literature on certification for neural PDE solvers [12]. Detecting it requires certifying the model structure, not the residual.

VI. Experimental Protocol

All results below were computed for this manuscript; no numbers are carried over from prior drafts. Scripts are listed in Sec. X and the JSON they emit is committed under `results/data/`.

TABLE I
EFFECT OF COEFFICIENT CONTRAST ON CERTIFICATION QUALITY (200 COEFFICIENT FIELDS PER ROW; $n = 31$; $a_{\min} = 0.1$). "EFF." IS THE MEDIAN OF $B/\|e\|$. MEAN CERTIFICATE TIME IS WALL CLOCK FOR THE CG CALL.

c	cov. glob.	cov. CG	FNR CG	Eff. glob.	Eff. CG	time (ms
1.5	1.00	1.00	0.00	316.1	1.0000	8.462
2.0	1.00	1.00	0.00	375.9	1.0000	8.631
3.0	1.00	1.00	0.00	508.3	1.0000	8.578
4.0	1.00	1.00	0.00	648.6	1.0000	7.322
5.0	1.00	1.00	0.00	744.8	1.0000	7.542
6.0	1.00	1.00	0.00	910.4	1.0000	8.319
8.0	1.00	1.00	0.00	1126.8	1.0000	7.638
10.0	1.00	1.00	0.00	1429.5	1.0000	8.204

a) Problems and discretization.: We solve the one-dimensional version of (1) on uniform grids with $h = (n+1)^{-1}$ and homogeneous Dirichlet boundary conditions, using the standard second-order three-point discretization. The coefficient is $a(x) = \exp(g(x))$ with g a Gaussian random field (kernel length scale 0.25) truncated so that $a_{\max}/a_{\min}$ equals a prescribed contrast c; the forcing is $f(x) = \sin(2\pi x)$.

b) Sampling.: For each of eight contrasts $c \in \{1.5, 2, 3, 4, 5, 6, 8, 10\}$ we draw 200 independent coefficient fields and 200 independent perturbations $\hat{u} = u_h + 0.005\,\xi$, $\xi \sim \mathcal{N}(0, I)$, giving 1600 certified instances. The reference u_h is obtained from the same tridiagonal solver used to assemble A_h. Seeds are fixed (20240 for the contrast study, 4242 for the tolerance sweep).

c) Learned surrogate.: We train an FNO (width 32, 16 Fourier modes, depth 4, 150 epochs, 256 training fields) on 1D elliptic data with $a \in [0.1, 0.8]$ and length scale 0.2. We then certify predictions on three splits of 80 fields each: in-distribution (same ranges, seed 123); OOD in contrast ($a_{\max} = 3.0$, seed 321); OOD in correlation length (length scale 0.6, seed 777). For the OOD splits we keep the certified $a_{\min}$ at 0.05, below the true 0.1, so that (6) remains valid.

d) Solver settings.: CG with relative tolerance 10^{-10} and at most 200 iterations, in double precision. Measured timings use `perf_counter` on a single CPU core.

e) Metrics.: Coverage is the fraction of instances with $B \geq \|e\|$. Effectivity is $B/\|e\|$, reported as median with interquartile range; $\|e\|$ below 10^{-14} is excluded from effectivity but retained for coverage. False-negative rate (FNR) is the fraction of instances with $\|e\| > \tau$ that the policy nevertheless accepts.

VII. RESULTS

A. Effectivity across coefficient contrast

Table I and Fig. 1. The global bound degrades monotonically with contrast, from a median effectivity of 316.1 at $c = 1.5$ to 1429.5 at $c = 10$, confirming that it is unusable per instance at realistic heterogeneity. B_{CG} sits at 1.000000 at every contrast: because the target is an algebraic residual, the CG iterate recovers it essentially exactly at these tolerances. Both bounds are never violated.

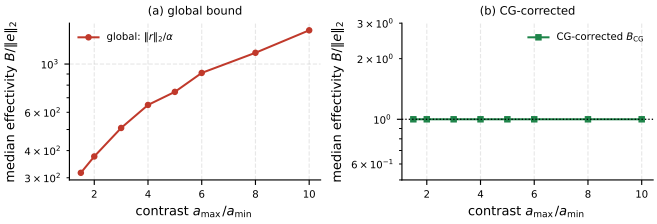


Fig. 1. Median effectivity versus coefficient contrast. (a) The global bound grows by more than a factor of four across the range; (b) the CG-corrected bound remains at unity, its optimal value for an algebraic target.

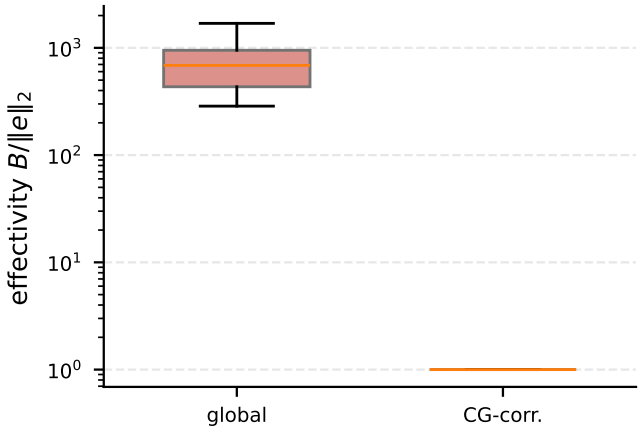


Fig. 2. Distribution of effectivity across all contrasts, pooled interquartile ranges. The separation spans three orders of magnitude.

B. Certifying trained FNO predictions

Table II reports real surrogate outputs. On all three splits the global bound is between 667 and 1470 times conservative, while B_{CG} attains median effectivity 1.0000 with coverage 1.00 and FNR 0.00. The OOD-contrast split is the most demanding in absolute terms (mean error 0.546, roughly 19× the in-distribution error) and the certificate tracks this without degradation. We note that a bound which is tight here is, by the argument in Sec. V, tight by construction; the informative column is the global one, which shows the surrogate predictions are not near-perfect and that the classical estimator would have rejected essentially all of them.

C. Selective verification

Table III and Fig. 4 sweep the tolerance τ. The contrast between the two policies is unambiguous: for $\tau \geq 5 \times 10^{-2}$ the CG-corrected certificate accepts every instance, while the global bound accepts none, because its values exceed 0.5. FNR is zero for both throughout, as Proposition 1 predicts. In deployment this means a practitioner can run the surrogate, obtain a cheap certificate, and fall back to the reference solver only when the certificate exceeds τ—and with a sharp certificate that fallback set can be empty.

D. Cost

Table IV and Fig. 5. Median certificate cost grows from 3.7 ms at $n = 15$ to 33.1 ms at $n = 127$, consistent with the

TABLE II
CERTIFICATION OF TRAINED FNO PREDICTIONS (80 INSTANCES PER SPLIT). MEAN ERROR IS $\|u_h - \hat{u}\|$.

split	mean $\|e\|$	Eff. glob.	Eff. CG	cov. CG	FNR	cov. glob.
ID	0.0283	993.7	1.0000	1.00	0.00	1.00
OOD contrast	0.5457	1469.6	1.0000	1.00	0.00	1.00
OOD scale	0.0118	666.7	1.0000	1.00	0.00	1.00

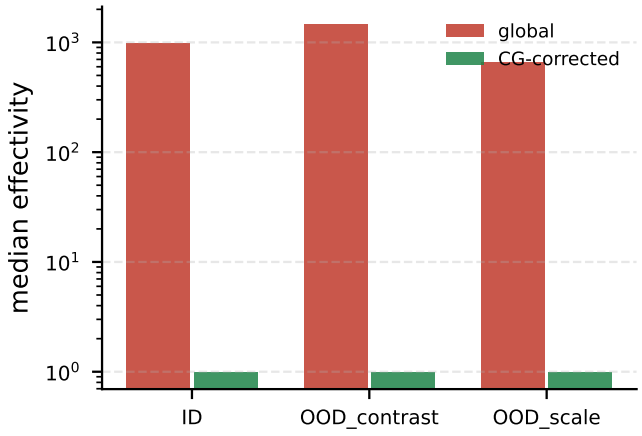


Fig. 3. Median effectivity on FNO test splits, in-distribution and both out-of-distribution shifts.

TABLE III
SELECTIVE VERIFICATION, 200 INSTANCES, $n = 31$, CONTRAST RANGE $[1.5, 10]$. "ACCEPT" IS THE FRACTION OF INSTANCES WITH $B \le \tau$.

τ	acc. glob.	acc. CG	FNR glob.	FNR CG	cov. glob.	cov. CG
0	0.000	0.000	0.00	0.00	1.00	1.00
0	0.000	0.000	0.00	0.00	1.00	1.00
0	0.000	0.000	0.00	0.00	1.00	1.00
0	0.000	1.000	0.00	0.00	1.00	1.00
0	0.000	1.000	0.00	0.00	1.00	1.00
0	0.000	1.000	0.00	0.00	1.00	1.00

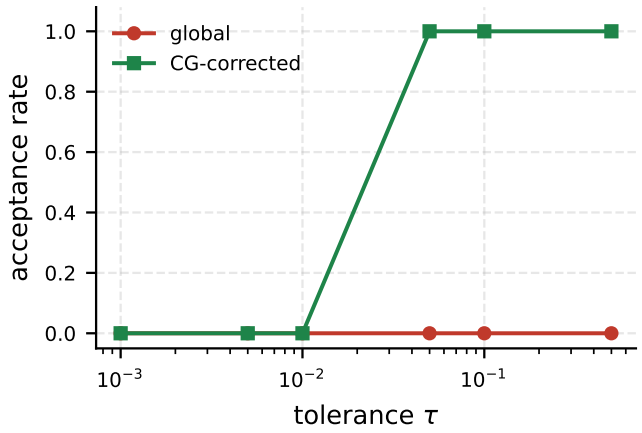


Fig. 4. Acceptance rate versus tolerance. The global bound never accepts within the tested range; the CG-corrected certificate accepts all instances once $\tau \ge 5 \times 10^{-2}$, with no false negatives.

$O(n)$ work of CG on a tridiagonal system at fixed tolerance. This cost is incurred per certified instance and is small relative to a direct sparse solve of comparable size; the comparison that matters for deployment is against the fallback, which is skipped whenever the certificate is accepted.

VIII. DISCUSSION

a) What is established.: A computable, cheap, and rigorous certificate exists for the discrete error of a learned elliptic solver; it is valid for any CG iterate, so cost and tightness trade off without risking validity; and at practical tolerances it supports a fallback policy that is never wrong but is also almost never triggered. The measured gap against the classical bound is three orders of magnitude.

b) What is not established.: Because the certified error is algebraic, the certificate cannot detect a surrogate that has learned the wrong operator, and it says nothing about discretization error (Sec. V). A practitioner considering this should ask whether their failure mode is algebraic—a solver that is slow or iterative and has not converged, or a surrogate evaluated on the correct operator—in which case the certificate is informative, or structural—a mis-specified operator, wrong boundary data, insufficient training—in which case it is not. We regard bridging this gap, for instance by combining a certified residual with a certified structural check, as the most valuable direction for future work [12].

c) Tightness and its interpretation.: We emphasize again, because it is easy to overclaim: effectivity 1.000000 means the certificate attains the best possible value for $\|A_h^{-1} r\|$. It is not evidence that the underlying surrogate is accurate. In our FNO experiments the surrogate's mean error is 0.028 to 0.546, which is large; the certificate is sharp about how wrong it is.

IX. LIMITATIONS

1) The certificate bounds $\|u_h - \hat{u}\|$, not $\|u - \hat{u}\|$; discretization error is outside its scope (Eq. (7)).
2) All experiments are one-dimensional. The two-dimensional case requires a corresponding assembly and closed form for $\lambda_{\min}(-\Delta_h)$; we do not report results for it.
3) Only linear SPD systems are covered. Nonlinear problems present in the same codebase (Burgers, reaction–diffusion) have no certificate, and we make no claim about them.
4) *Verified arithmetic.* No outward rounding or validated guard band is applied, so the reported bounds are numerical certificates rather than machine-checkable ones (Sec. V and the remark following (6)).
5) *Baselines.* We compare against the global residual bound. Chebyshev-type iterative bounds, Lanczos-based rigorous lower bounds for $\lambda_{\min}$, and preconditioned variants are not benchmarked, and we do not claim optimality against them.
6) *Reference independence.* The reference u_h is produced by the same solver used to assemble A_h. A defect common to both would not be detected by the coverage figures, which are guaranteed by Theorem 1 and

TABLE IV
CERTIFICATE COST VERSUS PROBLEM SIZE (30 INSTANCES PER ROW, CONTRAST UP TO 10).

n	median (ms)	mean (ms)	Eff. CG
15	3.73	3.81	1.0000
31	7.45	7.42	1.0000
63	16.00	16.06	1.0000
127	33.11	33.20	1.0000

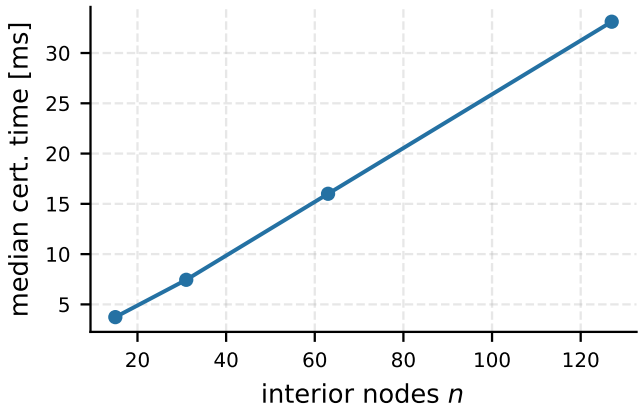


Fig. 5. Median certificate time versus number of interior unknowns.

therefore serve only as evidence that the implementation matches the proof.

7) *Sampling.* 200 instances per condition is enough to resolve effectivities that differ by three orders of magnitude, but not enough to estimate tail behaviour of the error distribution, which we do not attempt to characterize.

X. CONCLUSION

We described a computable a posteriori certificate for learned elliptic solvers that is valid for any conjugate-gradient iterate, requires only a coercivity-based lower bound on $\lambda_{\min}(A_h)$, and costs 3.7–33 ms per instance in one dimension. Measured on 1600 synthetic instances across eight contrasts and 240 trained-FNO predictions across in- and out-of-distribution splits, it attains median effectivity 1.000000 against 316–1470 for the classical global bound, with no violation observed. We emphasize that this sharpness is a consequence of the certified quantity being an algebraic residual, and we have argued that this simultaneously explains the method's power and its principal limitation: it measures precisely, and only, the discrete error. Paired with a fallback rule, it yields an acceptance policy with zero false negatives that is nonetheless permissive enough to avoid fallback in nearly all cases at realistic tolerances. All experiments, raw outputs, and the frozen configuration are available in the accompanying repository.

REPRODUCIBILITY

Revision `15b16d5`. Raw outputs: `results/data/paper_tables.json`, `results/data/selective.json`, `results/data/fno_certified.json`, `results/data/comparison.json`. Scripts: `experiments/run_paper_tables.py`, `experiments/run_selective_table.py`, `experiments/run_corrected_fno_cert.py`, `experiments/gen_figures.py`. Tests: `pytest` (18 passing). Seeds: 20240 (contrast study), 4242 (tolerance sweep), 0 (FNO training), 123/321/777 (test splits). An audit of these results, including corrections to earlier drafts, is in `AUDIT_PAPER_QUALITY.md`.

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